

Predicting Regional Poverty Levels in Bangladesh Using Satellite Night-Light Data and Geospatial Features

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Abstract

Accurate poverty measurement at fine spatial scales is critical for effective policy targeting, yet traditional household surveys are expensive, infrequent, and spatially coarse. This study develops a machine learning framework to predict district-level poverty in Bangladesh using freely available satellite and geospatial data, without requiring ground-level survey data at the district level. We extract 19 features from multiple remote sensing sources — including VIIRS nighttime light radiance, MODIS vegetation indices, ESA land cover, SRTM elevation, WorldPop population density, and OpenStreetMap road networks — for all 64 districts of Bangladesh. A Random Forest regressor, evaluated under Leave-One-Division-Out spatial cross-validation, achieves a root mean squared error of 3.626 percentage points against HIES 2022 poverty headcount ratios. Spatial autocorrelation analysis reveals strong poverty clustering (Moran’s $I = 0.733$, $p < 0.001$), motivating the inclusion of spatial lag features which contribute 29% of total model importance. Notably, the spatial lag of nighttime light (17.3% importance) outperforms the district’s own nighttime light (2.3%), indicating that geographic context is more predictive than individual satellite measurements. A transfer-learned ResNet-18 CNN trained on 4-band Sentinel-2 imagery achieves competitive mean absolute error (3.19 pp) but higher RMSE (4.35 pp), highlighting the limitations of deep learning with small spatial datasets. These findings demonstrate the feasibility of satellite-based poverty mapping in South Asian contexts and provide an open-source, reproducible pipeline for poverty monitoring at scale.

Keywords: poverty mapping, nighttime light, random forest, spatial autocorrelation, Bangladesh, remote sensing, machine learning, Sentinel-2

1. Introduction

Poverty measurement in developing countries relies predominantly on household survey data, which are costly to collect, infrequent in temporal coverage, and rarely available at sub-national spatial resolution (1). In Bangladesh, the most recent nationally representative Household Income and Expenditure Survey (HIES 2022) provides poverty estimates only at the division level — eight administrative units covering a population of 170 million (2). This spatial coarseness severely limits the ability of policymakers and development organizations to target interventions at the district or sub-district level.

Satellite remote sensing offers a cost-effective and temporally consistent alternative for poverty proxy measurement. Nighttime light (NTL) radiance, captured by the Visible Infrared Imaging Radiometer Suite (VIIRS), has been demonstrated as a robust proxy for

economic activity at national and sub-national scales (3; 4). More recently, daytime satellite imagery combined with convolutional neural networks (CNNs) has shown remarkable predictive power for wealth estimation at the village level in sub-Saharan Africa and South Asia (1).

Bangladesh presents a particularly important test case for satellite-based poverty mapping. As one of the most densely populated countries in the world, with a complex geography spanning coastal deltas, haor wetlands, and the Chittagong Hill Tracts, the relationship between observable satellite signals and poverty outcomes is likely to be geographically heterogeneous. Furthermore, the country has experienced rapid poverty reduction over the past decade — from 31.5% in 2010 to 18.7% in 2022 — making temporal monitoring of poverty patterns particularly valuable.

This study makes three principal contributions. First, we develop and evaluate a multi-source geospatial feature extraction pipeline covering nine satellite and geospatial datasets for all 64 districts of Bangladesh. Second, we demonstrate that spatial lag features — capturing the satellite characteristics of neighboring districts — substantially improve poverty prediction over non-spatial baselines. Third, we provide a direct comparison of tabular machine learning (Random Forest) against transfer-learned deep learning (ResNet-18 CNN) under a strict spatial cross-validation protocol, contributing to the methodological literature on poverty mapping at small sample sizes.

2. Related Work

2.1. Nighttime Light as an Economic Proxy

The use of satellite-derived nighttime light (NTL) as a proxy for economic activity has a well-established foundation in the literature. Henderson et al. (3) demonstrated that NTL data from DMSP-OLS satellites could augment official GDP estimates, particularly in countries with weak statistical infrastructure, by showing strong correlations between light growth and economic growth at sub-national scales. This work established NTL as a credible instrument for measuring economic output where ground-based data is sparse or unreliable.

Engstrom et al. (4) extended this approach by combining high-resolution daytime satellite imagery with object-level features — building density, roof materials, road length, and agricultural patterns — to estimate poverty rates and consumption expenditure across 1,291 administrative units in Sri Lanka. Their linear regression framework explained 49–61% of the variation in average consumption, demonstrating that visible-band imagery contains substantial poverty-relevant information beyond what NTL alone captures.

2.2. Machine Learning and CNN-Based Poverty Mapping

The landmark study by Jean et al. (1) demonstrated that a convolutional neural network (CNN) trained on high-resolution daytime satellite imagery, using nighttime light as a transfer learning signal, could predict consumption expenditure and asset wealth with high accuracy across five African countries. Their method explained up to 75% of the variation in local economic outcomes using only publicly available satellite data. This work fundamentally shifted the paradigm of poverty mapping, showing that deep learning on satellite imagery could substitute for expensive household surveys in data-scarce settings.

Ayush et al. (5) further advanced this direction by introducing a reinforcement learning framework that dynamically selects where to acquire costly high-resolution imagery based on cheap low-resolution inputs. Applied to poverty prediction in Uganda, their approach matched previous CNN benchmarks while requiring 80% fewer high-resolution images, addressing the practical cost barrier to scaling satellite-based poverty mapping.

2.3. Spatial Autocorrelation in Poverty Research

The spatial structure of poverty has long been recognized as a critical dimension of its measurement and explanation. Anselin (6) formalized the concept of Local Indicators of Spatial Association (LISA), providing a statistical framework for detecting and quantifying spatial clustering patterns. Poverty in developing countries consistently exhibits strong positive spatial autocorrelation — poor communities cluster geographically due to shared agro-ecological conditions, infrastructure deficits, and market access constraints.

Despite this well-known spatial structure, most satellite-based poverty mapping studies treat observations as spatially independent, failing to incorporate neighborhood context as a predictive feature. The PySAL library (7) provides robust tools for spatial analysis in Python, enabling the computation of spatial weights matrices and spatial lag variables that capture this neighborhood dependency. The present study addresses this gap by explicitly constructing and incorporating spatial lag features, motivated by Moran’s I analysis of Bangladesh’s poverty distribution.

2.4. Gap and Contribution

Existing satellite-based poverty mapping studies have primarily focused on sub-Saharan Africa and South-East Asia, with relatively limited attention to Bangladesh and the broader South Asian delta context (1; 4). Furthermore, few studies have systematically compared tabular machine learning against CNN transfer learning under strict spatial cross-validation protocols, or have explicitly quantified the contribution of spatial lag features relative to individual district measurements. This study directly addresses both gaps by developing a spatially-aware, multi-source feature extraction pipeline for Bangladesh and providing a rigorous methodological comparison across model families under Leave-One-Division-Out evaluation.

3. Study Area

Bangladesh is a South Asian country bordered by India and Myanmar, with a land area of approximately 147,570 km² and a population of approximately 170 million (2022 estimate). The country is administratively divided into 8 divisions, 64 districts, and 495 upazilas. This study uses the 64 districts as the primary spatial unit of analysis.

The geography of Bangladesh is dominated by the Ganges-Brahmaputra-Meghna (GBM) delta system, which covers approximately 80% of the country. The remaining 20% comprises the Chittagong Hill Tracts in the southeast — a mountainous forested region with elevations exceeding 900m — and the Madhupur Tract in the north-centre. This geographic diversity produces substantial spatial heterogeneity in poverty outcomes, land use, and economic activity.

Poverty rates across Bangladesh’s eight divisions range from 14.8% (Khulna) to 26.9% (Barishal) under the upper poverty line, with Rangpur (24.8%) and Mymensingh (24.2%) also exhibiting elevated poverty rates (HIES 2022). The geographic distribution of poverty shows a clear north-south pattern, with northern and southern peripheral divisions experiencing higher poverty than the central Dhaka-Chattogram corridor.

4. Data and Methods

4.1. Ground Truth Data

Poverty headcount ratios (HCR) were obtained from the Bangladesh Household Income and Expenditure Survey 2022 (HIES 2022), published by the Bangladesh Bureau of Statistics (BBS). The upper poverty line HCR represents the proportion of the population whose per capita consumption expenditure falls below the upper poverty line. As HIES 2022 reports poverty at the division level, all districts within a division were assigned the same poverty label — a known limitation discussed in Section 5.

4.2. Satellite and Geospatial Features

All satellite features were extracted using Google Earth Engine (GEE) (8), with OpenStreetMap road data extracted using the OSMnx Python library (9). Features were aggregated to the district level using the `reduceRegions()` function in GEE with Queen contiguity boundaries from GADM 4.1.

4.2.1. Nighttime Light (NTL)

Monthly NTL composites from VIIRS Day/Night Band (NOAA/VIIRS/DNB/MONTHLY_V1/VCMC) were extracted for 2018–2022. Annual median composites were computed for each year and the following statistics extracted per district: mean, standard deviation, maximum, minimum, 25th and 75th percentiles. Three temporal features were derived: year-on-year change (2021–2022), linear trend slope (2018–2022), and interquartile range (IQR) as a proxy for within-district inequality.

4.2.2. Vegetation Index

Mean and standard deviation of the Normalized Difference Vegetation Index (NDVI) were extracted from MODIS Terra MOD13A3 monthly composites for 2022. NDVI serves as a proxy for agricultural activity and rural land use.

4.2.3. Land Cover

Urban fraction (proportion of district classified as urban land, class 50) and water fraction (class 80) were extracted from ESA WorldCover v200 (2021) at 10m resolution.

4.2.4. Elevation

Mean and standard deviation of elevation were extracted from the SRTM Digital Elevation Model at 30m resolution.

4.2.5. Population Density

Mean and total population density were extracted from the WorldPop Global Population Density dataset (2020) at 100m resolution.

131 4.2.6. Road Network

132 Total road length and road density (km per km²) were calculated by downloading drive-
133 able road networks from OpenStreetMap for each district using OSMnx.

134 4.3. Feature Engineering

135 Four derived features were computed. NTL per capita was calculated as the district
136 mean NTL divided by population density, normalizing economic activity for population size.
137 Spatial lag features were computed for NTL mean, NTL per capita, NDVI mean, and pop-
138 ulation density using row-standardized Queen contiguity spatial weights, following Anselin
139 (6). These spatial lag features capture the average satellite characteristics of neighboring
140 districts.

141 4.4. Spatial Autocorrelation Analysis

142 Global Moran’s I was computed for poverty HCR and key satellite features using Queen
143 contiguity weights and 999 random permutations for significance testing, implemented using
144 the PySAL library (7).

145 4.5. Machine Learning Models

146 4.5.1. Baseline Models

147 Linear Regression, Ridge Regression, and Lasso Regression were trained as baseline mod-
148 els. All three failed to outperform the naive mean predictor under spatial cross-validation,
149 primarily due to extrapolation failure on Dhaka district’s extreme feature values.

150 4.5.2. Random Forest

151 A Random Forest regressor (10) was trained on 19 features. Hyperparameters were
152 tuned using GridSearchCV with Leave-One-District-Out cross-validation. Optimal param-
153 eters were: `max_depth=None`, `max_features=0.7`, `min_samples_leaf=1`, `n_estimators=100`.
154 Feature scaling used RobustScaler to mitigate the influence of outliers.

155 4.5.3. CNN with Transfer Learning

156 A ResNet-18 (11) pretrained on ImageNet was fine-tuned for poverty regression. The
157 first convolutional layer was modified to accept 4-band Sentinel-2 input (RGB + NIR at
158 100m resolution), with NIR weights initialized from the mean of the RGB channels. The
159 final fully connected layer was replaced with a regression head (Linear(512, 64) → ReLU →
160 Dropout(0.2) → Linear(64, 1)). Images were resampled to 224×224 pixels. Training used
161 Adam optimizer with MSE loss and ReduceLROnPlateau scheduler.

162 4.6. Evaluation Protocol

163 All models were evaluated using Leave-One-Division-Out (LODO) cross-validation: in
164 each fold, one division’s districts were held out for testing and the model trained on the
165 remaining seven divisions’ districts, rotated across all eight divisions. This spatial blocking
166 prevents information leakage from neighboring districts and provides a conservative, realistic
167 estimate of out-of-sample performance. Metrics reported are RMSE, MAE, and R² averaged
168 across all 64 test districts.

5. Results

5.1. Spatial Autocorrelation

All five variables tested exhibited statistically significant positive spatial autocorrelation (Table 1). Poverty HCR showed the strongest clustering ($I = 0.733$, $p < 0.001$), indicating that poverty is highly geographically concentrated in Bangladesh. NTL per capita ($I = 0.460$) and NDVI mean ($I = 0.469$) also showed moderate clustering, validating the use of spatial lag features.

Table 1: Global Moran’s I spatial autocorrelation test results (Queen contiguity weights, 999 permutations).

Variable	Moran’s I	p-value	Significant
Poverty HCR	0.733	0.001	Yes ***
NTL Mean	0.320	0.007	Yes ***
NTL per Capita	0.460	0.001	Yes ***
NDVI Mean	0.469	0.001	Yes ***
Population Density	0.253	0.001	Yes ***

5.2. Model Performance

Table 2 presents LODO cross-validation performance for all models. The Random Forest model (RMSE = 3.626 pp) outperforms the naive mean predictor baseline (RMSE = 4.163 pp) by 12.9%, representing a meaningful improvement given the constraint of eight unique target values. Linear models failed entirely under spatial cross-validation (RMSE 16–28 pp), confirming the non-linear relationship between satellite features and poverty. The CNN achieved comparable MAE to Random Forest (3.19 vs 2.93 pp) but higher RMSE (4.35 pp), indicating more frequent large errors on out-of-distribution divisions.

Table 2: Model performance under Leave-One-Division-Out cross-validation (64 districts).

Model	RMSE (pp)	MAE (pp)	R ²
Naive Baseline	4.163	3.596	0.000
Linear Regression	28.203	10.451	-44.900
Ridge Regression	20.707	8.907	-23.745
Lasso Regression	16.458	7.965	-14.631
XGBoost	6.104	5.112	-1.150
LightGBM	5.679	4.768	-0.861
CNN ResNet-18	4.354	3.188	-0.094
Random Forest	3.626	2.926	0.241

5.3. Feature Importance

The two most important features were elevation mean (19.2%) and NTL mean spatial lag (17.3%), as shown in Figure 1. Collectively, spatial lag features contributed 28.9% of total importance, confirming that geographic context is more predictive than individual district measurements. The district’s own NTL mean contributed only 2.3%, substantially less than its spatial lag counterpart (17.3%).

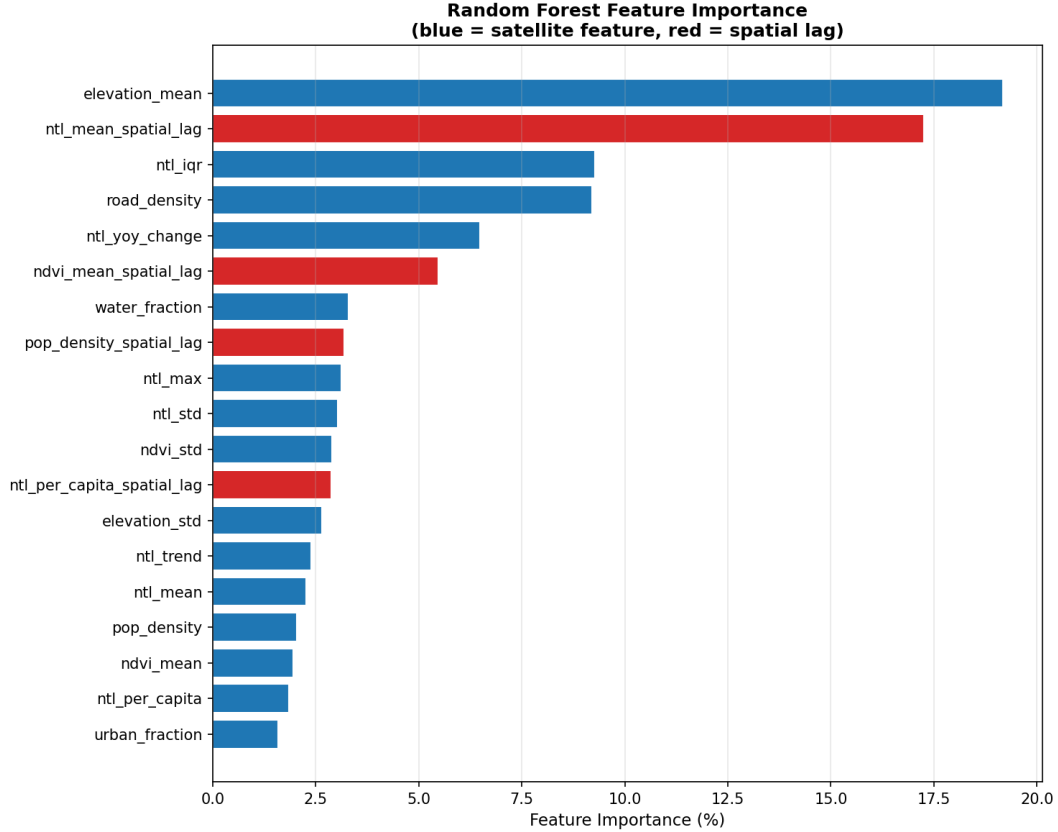


Figure 1: Random Forest feature importance (%). Red bars indicate spatial lag features; blue bars indicate individual satellite features. Spatial lag features collectively contribute 28.9% of total importance.

5.4. Spatial Error Analysis

The Random Forest model systematically underpredicts poverty in Barishal division (true HCR: 26.9%, mean predicted: 21.6%) and Rangpur division (true: 24.8%, mean predicted: 20.7%). Conversely, Dhaka division is predicted with high accuracy (MAE = 0.47 pp), likely due to its distinctive satellite feature profile. This spatial bias pattern is discussed further in Section 6.

6. Discussion

6.1. The Role of Spatial Context

The finding that the spatial lag of NTL (17.3% importance) substantially outperforms the district's own NTL (2.3%) has important methodological implications for poverty mapping research. It suggests that the economic neighborhood of a district — its connectivity to urban centres, regional market access, and proximity to economic corridors — is a stronger determinant of poverty than the district's own observed luminosity. This is consistent with spatial economics theory (6) and with the strong Moran's I values observed for poverty ($I = 0.733$, $p < 0.001$).

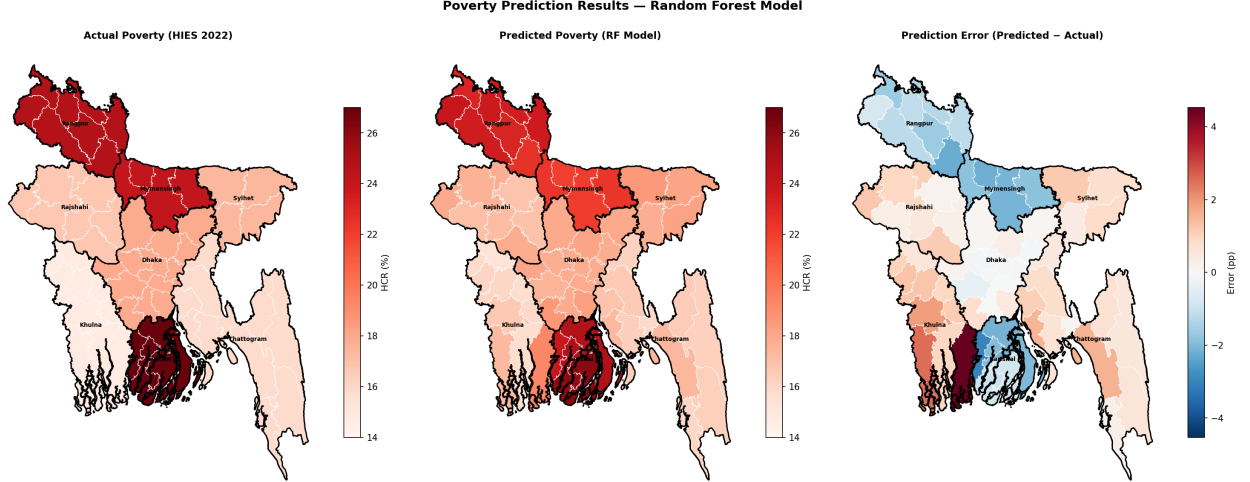


Figure 2: Left: Actual poverty headcount ratio by district (HIES 2022). Centre: Random Forest model predictions. Right: Prediction error (predicted – actual). Division boundaries shown in black.

6.2. Deep Learning vs Tabular Machine Learning

Despite the theoretical advantage of CNNs in capturing spatial texture patterns invisible to tabular features (1), the Random Forest outperformed ResNet-18 on RMSE under strict spatial cross-validation. This result is consistent with Jean et al. (1), who note that transfer learning from nighttime light to daytime imagery requires large labeled datasets — their study used over 10,000 labeled survey clusters compared to our 64 districts. At very small sample sizes, the inductive bias of tree-based ensembles appears to provide superior generalization.

6.3. Limitations

Several limitations must be acknowledged. First, the use of division-level poverty labels for district-level prediction introduces label noise — 13 Dhaka districts with vastly different satellite characteristics share the same poverty label of 17.9%. This fundamentally limits model discriminability within divisions and is the primary constraint on achievable R^2 . Second, the small sample size ($n = 64$) limits statistical power and makes cross-validation estimates high-variance. Third, the satellite features capture economic activity proxies rather than consumption poverty directly; the causal pathway from, for example, elevation to poverty operates through multiple unobserved mediators.

6.4. Policy Implications

Despite these limitations, the framework demonstrates that satellite-based poverty monitoring can provide useful spatial signal for resource allocation decisions. Districts like Barishal and Rangpur — which the model identifies as high-poverty with characteristic satellite signatures (low NTL, high water fraction, high vegetation) — could be prioritized for targeted development interventions without requiring expensive surveys. The open-source pipeline developed here is directly applicable to other South Asian contexts where sub-national poverty data is sparse.

7. Conclusion

This study demonstrates that satellite nighttime light, vegetation indices, terrain data, and road network features can predict district-level poverty headcount ratios in Bangladesh with a RMSE of 3.626 percentage points under strict spatial cross-validation. The key finding is that spatial context — captured through the spatial lag of neighboring districts’ satellite features — contributes nearly 29% of total model importance, substantially exceeding the predictive power of individual district measurements. This underscores the importance of incorporating spatial autocorrelation structure into satellite-based poverty mapping frameworks.

The comparison between Random Forest and CNN-based approaches reveals a fundamental constraint: transfer learning from large image datasets does not compensate for the absence of fine-grained poverty labels at the district level. Future work should prioritize obtaining sub-division poverty estimates — ideally at the upazila level — which would enable both improved model training and more actionable policy targeting. Integration with mobile phone call detail records and financial transaction data may further improve prediction accuracy.

The complete codebase, trained models, and extracted satellite features are publicly available at https://github.com/raihan-ahmed-khan/Poverty_Prediction_BD.

References

- [1] N. Jean, M. Burke, M. Xie, W. M. Davis, D. B. Lobell, S. Ermon, Combining satellite imagery and machine learning to predict poverty, *Science* 353 (6301) (2016) 790–794. doi:10.1126/science.aaf7894. URL <https://www.science.org/doi/10.1126/science.aaf7894>
- [2] Bangladesh Bureau of Statistics, Household income and expenditure survey 2022, Tech. rep., Bangladesh Bureau of Statistics, Dhaka, Bangladesh (2022).
- [3] J. V. Henderson, A. Storeygard, D. N. Weil, Measuring Economic Growth from Outer Space, *American Economic Review* 102 (2) (2012) 994–1028. doi:10.1257/aer.102.2.994. URL <https://www.aeaweb.org/articles?id=10.1257/aer.102.2.994>
- [4] R. Engstrom, J. Hersh, D. Newhouse, Poverty from Space: Using High Resolution Satellite Imagery for Estimating Economic Well-being, *The World Bank Economic Review* 36 (2) (2022) 382–412. doi:10.1093/wber/lhab015. URL <https://doi.org/10.1093/wber/lhab015>
- [5] K. Ayush, B. Uz kent, K. Tanmay, M. Burke, D. Lobell, S. Ermon, Efficient Poverty Mapping from High Resolution Remote Sensing Images, *Proceedings of the AAAI Conference on Artificial Intelligence* 35 (1) (2021) 12–20. doi:10.1609/aaai.v35i1.16072. URL <https://ojs.aaai.org/index.php/AAAI/article/view/16072>
- [6] L. Anselin, Local Indicators of Spatial Association—LISA, *Geographical Analysis* 27 (2) (1995) 93–115. doi:10.1111/j.1538-4632.1995.tb00338.x.

268 URL [https://onlinelibrary.wiley.com/doi/10.1111/j.1538-4632.1995.](https://onlinelibrary.wiley.com/doi/10.1111/j.1538-4632.1995.tb00338.x)
269 [tb00338.x](https://onlinelibrary.wiley.com/doi/10.1111/j.1538-4632.1995.tb00338.x)

270 [7] S. J. Rey, L. Anselin, PySAL: A python library of spatial analytical methods, *The*
271 *Review of Regional Studies* 37 (1) (2007) 5–27.

272 [8] N. Gorelick, M. Hancher, M. Dixon, S. Ilyushchenko, D. Thau, R. Moore, Google earth
273 engine: Planetary-scale geospatial analysis for everyone, *Remote Sensing of Environment*
274 202 (2017) 18–27.

275 [9] G. Boeing, OSMnx: New methods for acquiring, constructing, analyzing, and visualizing
276 complex street networks, *Computers, Environment and Urban Systems* 65 (2017) 126–
277 139.

278 [10] L. Breiman, Random Forests, *Machine Learning* 45 (1) (2001) 5–32. doi:10.1023/A:
279 1010933404324.
280 URL <https://link.springer.com/10.1023/A:1010933404324>

281 [11] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, in:
282 *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016,
283 pp. 770–778.
284 URL [http://openaccess.thecvf.com/content_cvpr_2016/html/He_Deep_](http://openaccess.thecvf.com/content_cvpr_2016/html/He_Deep_Residual_Learning_CVPR_2016_paper.html)
285 [Residual_Learning_CVPR_2016_paper.html](http://openaccess.thecvf.com/content_cvpr_2016/html/He_Deep_Residual_Learning_CVPR_2016_paper.html)